SCIENTIFIC RESEARCH

NKTRIF - A MULTIMODAL STABLE FUSION APPROACH FOR FINE-GRAINED STOCK MOVEMENT PREDICTION

Instructor

Ph.D. Le Dien Tuan

M.Sc Chau Thi Da Huong

Authors

Tieu Thanh Mai Anh

Nguyen Viet Khoa

Ngo Tan Phong

Le Hung Phuc

Tran Vu Ha My

RESEARCH OUTLINES

01 INTRODUCTION

02 RELATED WORKS

03 METHODOLOGY

04 RESULTS

05 CONCLUSION AND FUTURE WORK

1. Research Background & Problems

Stock movement prediction is a central problem in **quantitative finance**, with direct implications for **asset allocation, trading strategy** design, and **risk management**.

The stock market is a complex system driven by multiple information sources:

Technical Price Indicators

Reflect historical price dynamics (Close, Open, High)

Event-Driven Financial News

Captures corporate events, investor expectations, and sector-level movements

Macroeconomic Indices

Reflect system-level market conditions

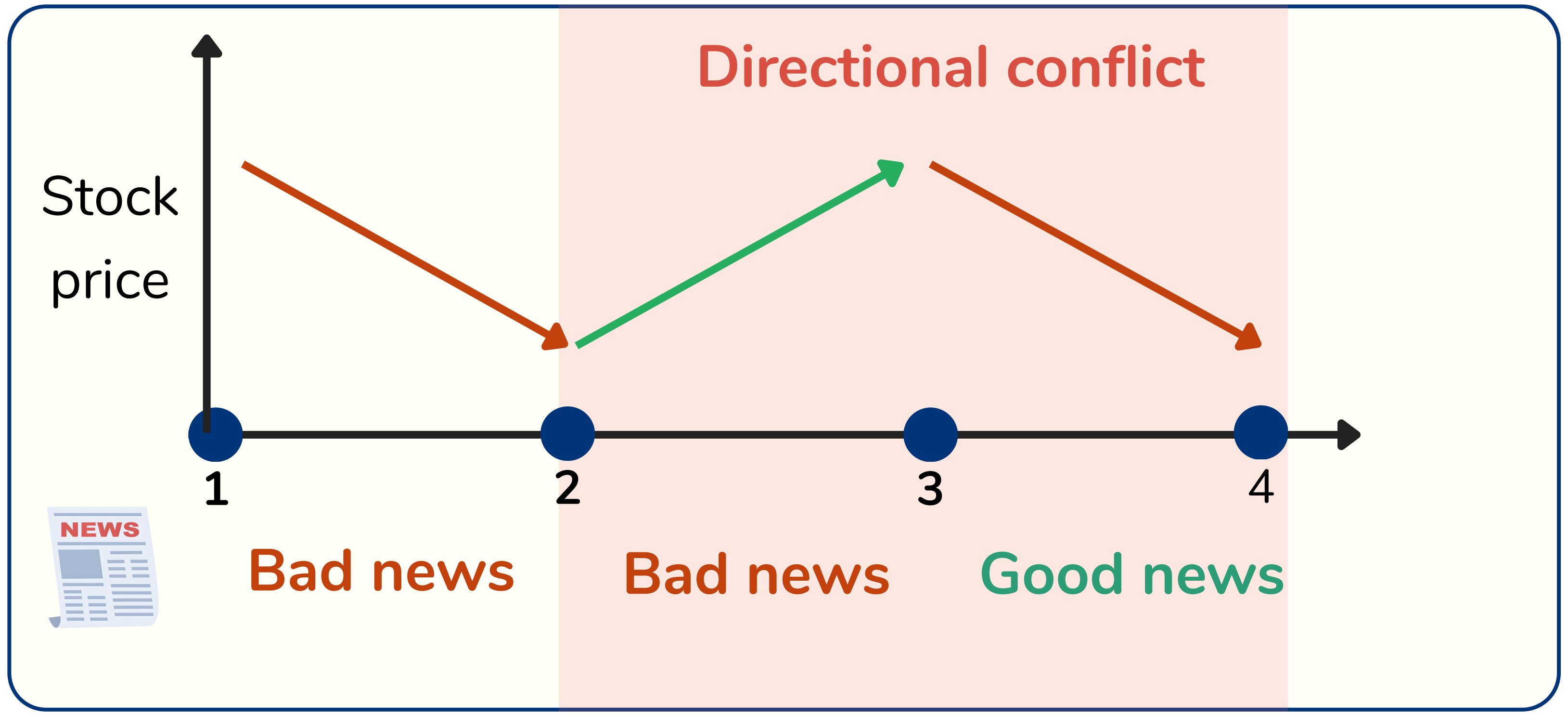
Challenge: **Designing a fusion mechanism that respects the characteristics of each source.**

1. Research Background

Development of forecasting methods

TIME SERIES MODEL	TEXT-BASED MODEL	MULTIMODAL FUSION
Statistical autocorrelation, non-linear recurrence, and global self-attention to encode sequential price dynamics (ex: ARIMA)	Static vectors, domain-adapted contextual PLMs, and structured LLM prompting to isolate high-density financial signals (ex: FinBERT)	Concatenation, bilinear pooling, raw co-attention to fuse many information sources (ex: ESTIMATE using Multihead Attention)

2. Research gap



Mention Company X
Mention Company Y
Mention Company Z

01 Cross-modal semantic conflict

Occurs when market signals deliver **contradictory** or **time-lagged** directional indicators simultaneously

02 Sparsity of financial news

News clusters around discrete events, leaving many trading day news-empty.

03 Traditional embedding approach

News raw-text embedding can not distinct semantic effect between multiple companies and are affected by linguistic noises.

3. Research objectives

Theoretical Objective

Design **NKTriF**, a novel architecture utilizing a **two-stage Gated Cross-Attention** module to utilize price series, S-R-O news triples, and macro indices for financial short-term trend prediction

Empirical Objective

Evaluate performance on S&P 500 constituents with fine-grained **3-class prediction** to verify and confim the **advance** and **stabling ability** of proposed models to a **variety of information types**.

Application Objective

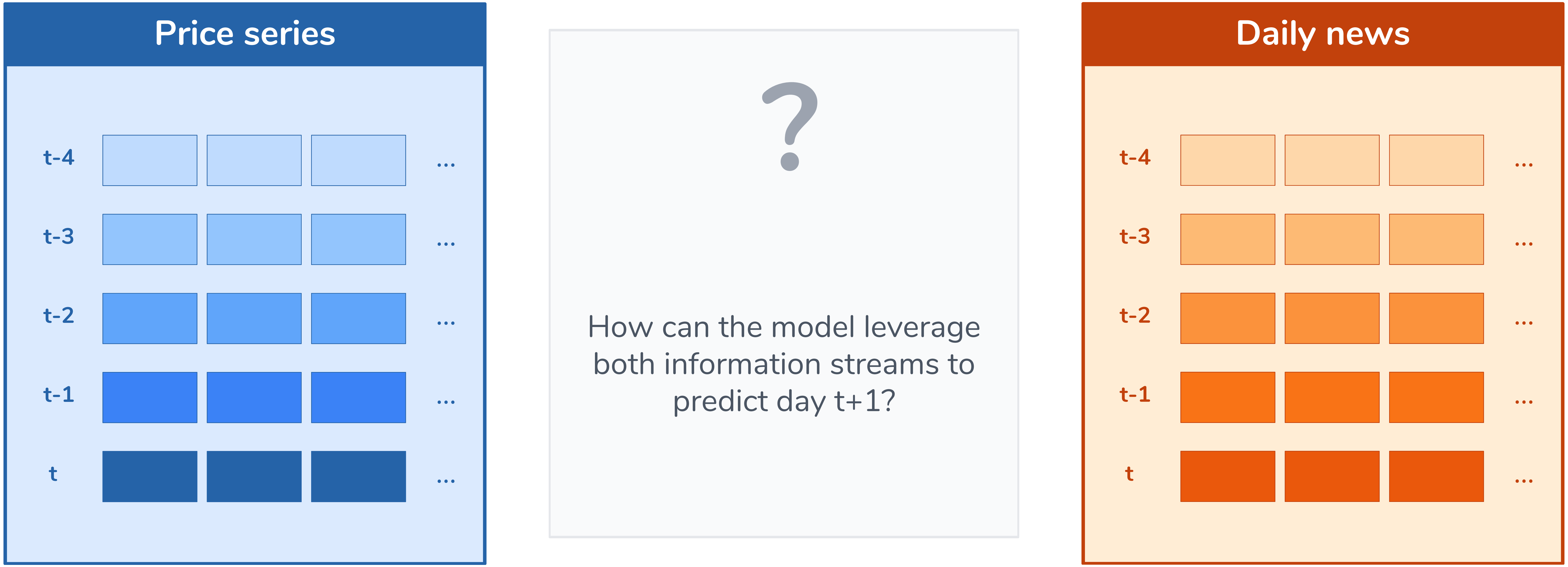
Construct an **end-to-end pipeline** featuring LLM-driven structured knowledge extraction paired with FinBERT semantic encoding.

LITERATURE REVIEW: BASELINE TAXONOMY & PERFORMENCE MATRIX

Method	Author (Year)	Historical Price	Finanacal News	Macro Indicators	Prediction	Fusion mechanism	ACC
LSTM	Fischer & Krauss (2018)	✓	-	-	2 class	Temporal gating	**N/A**
ALSTM	Hu et al. (2018)	✓	-	-	2 class	Temporal attention	**0.4869 - 0.5254**
DA-RNN	Qin et al. (2017)	✓	-	-	2 class	Dual-stage attention	**N/A**
DTML	Yoo et al. (2021)	✓	-	✓	2 class	Multi-level Transformer	**0.5276 - 0.5744**
ESTIMATE	Huynh et al. (2023)	✓	-	✓	2 class	Concat + MHA	**N/A**
StockNet	Xu & Cohen (2018)	✓	✓	-	2 class	VAE	**0.5235 - 0.5360**
SLOT	Soun et al. (2022)	✓	✓	-	2 class	Cross-attention	**0.5481 - 0.5872**
MSGCA	Zong et al. (2025)	✓	-	✓	**3 class**	Gated cross-attention	**0.3861 - 0.4379**

GAP: Existing literature lacks intergrated architectures that concurrently **Price + News + Macro** signals under fine-grained 3-class constrains

Problem: How can we use both Stock prices and News information to predict stock movement?



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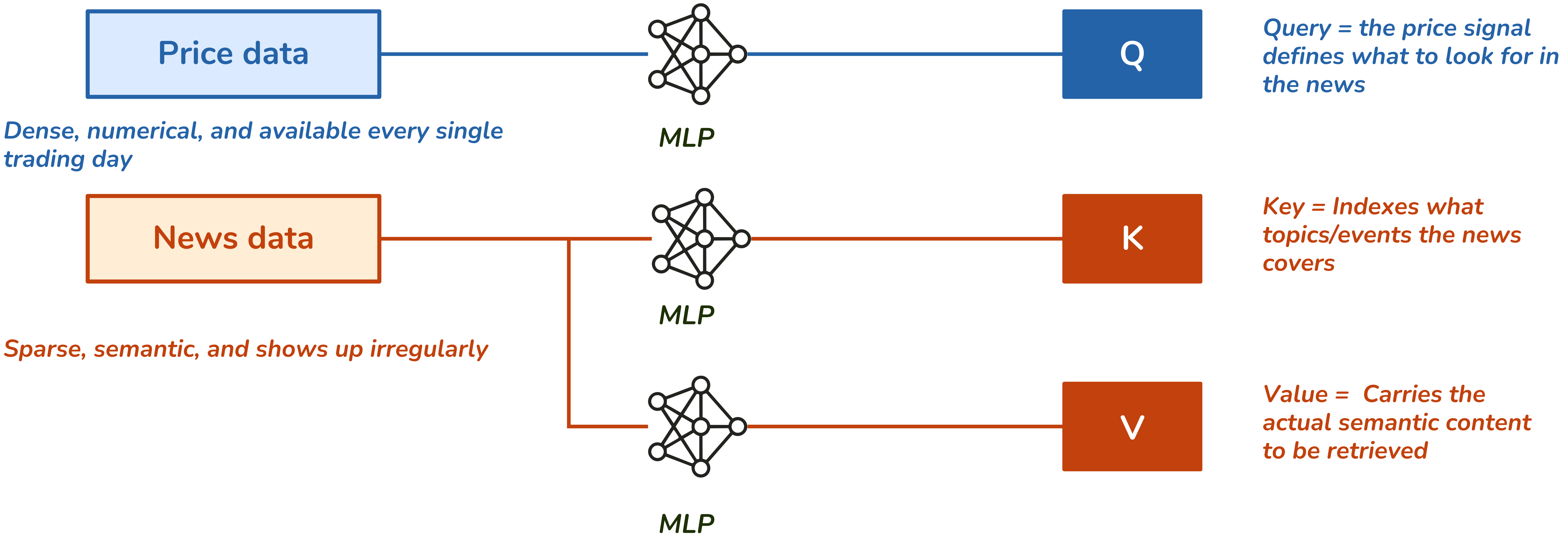
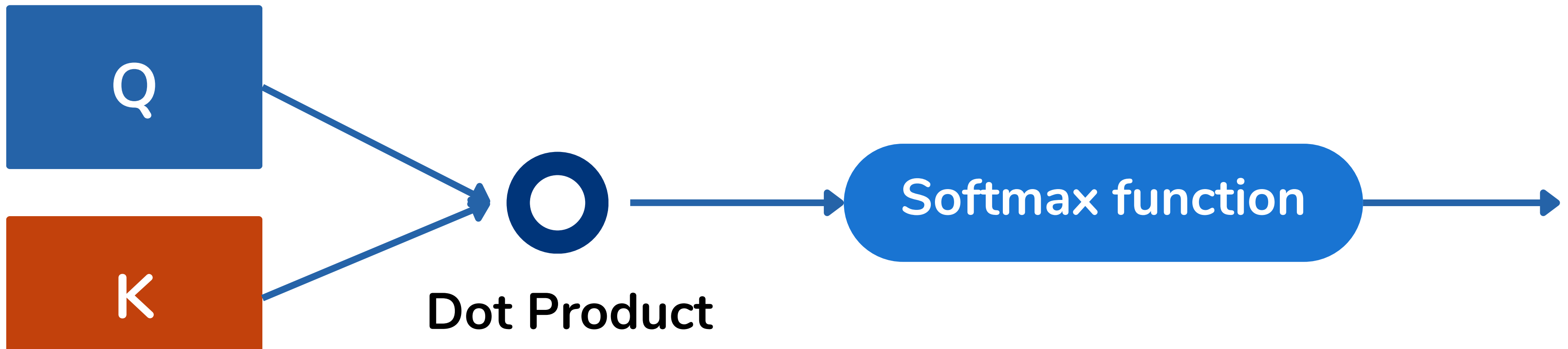


By separating Price → Q and News → K, V, the model forces price dynamics to ask the questions, while news answers them

Attention matrix: Which information is attended to?



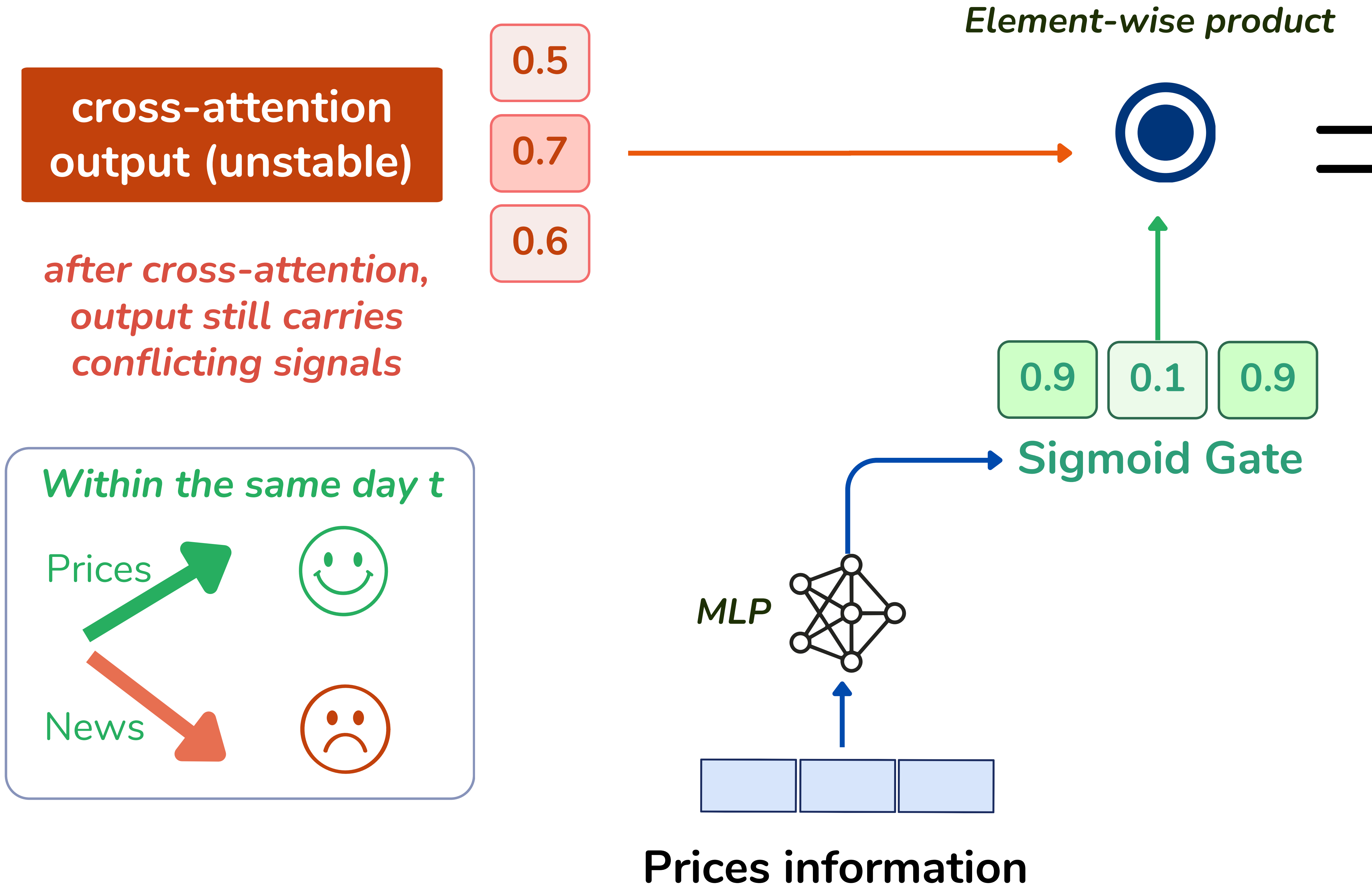
Attention matrix between Q - K

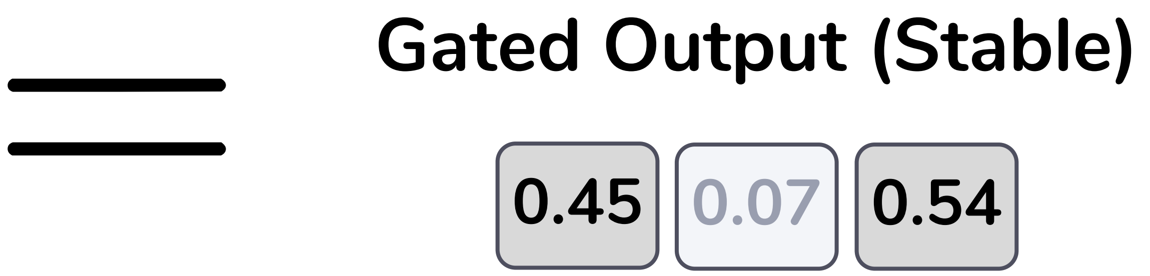
	t-5	t-4	t-3	t-2	t-1	t
t-5	0.38	0.22	0.18	0.12	0.06	0.04
t-4	0.10	0.42	0.26	0.14	0.05	0.03
t-3	0.05	0.12	0.45	0.28	0.07	0.03
t-2	0.03	0.06	0.14	0.52	0.20	0.05
t-1	0.02	0.04	0.07	0.18	0.55	0.14
t	0.01	0.02	0.04	0.08	0.20	0.65

The attention matrix captures the complex interaction between two information streams.

Fused output: weighted sum of news value at each price timesteps

Gated Stable Feature Selection: Controlling noise after Cross-Attention



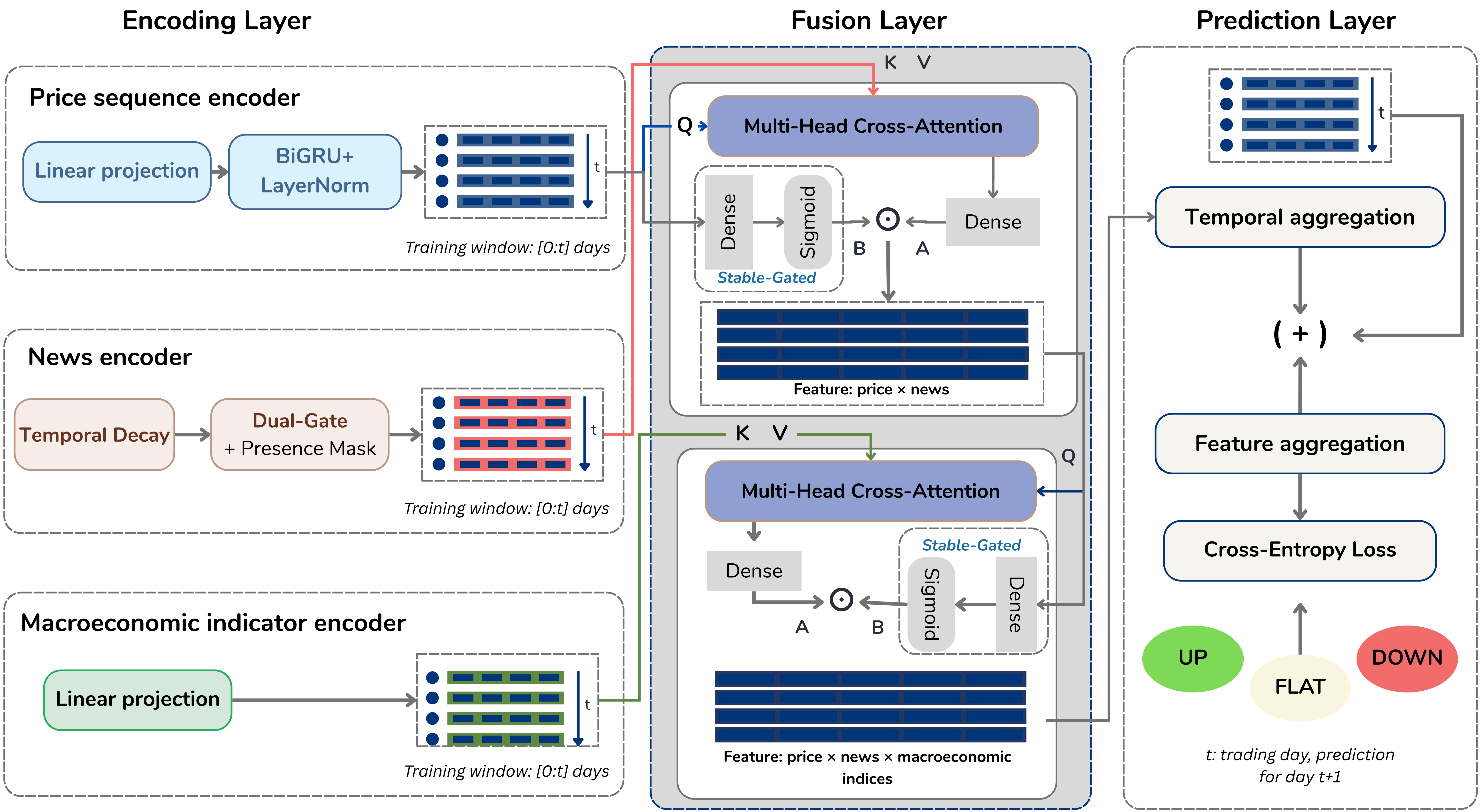


A **price-conditioned sigmoid gate** learns which dimensions to trust:

- Gate ≈ 1 → signal is **consistent** with price context → **pass through**
- Gate ≈ 0 → signal **conflicts** with price → **suppress**

Price is chosen as stablizing source by its consistency and its direct reflection of financial market.

Dauphin et al. (2017) - Language modeling with gated convolutional networks


Encoding Layer
Fusion Layer
Prediction Layer
Price sequence encoder
Linear projection
BiGRU+ LayerNorm
t
Training window: [0:t] days
News encoder
Temporal Decay
Dual-Gate + Presence Mask
Training window: [0:t] days
Macroeconomic indicator encoder
Linear projection
Training window: [0:t] days
K V
Q
Multi-Head Cross-Attention
Dense
Sigmoid
Stable-Gated
Dense
B
A
Feature: price × news
K V
Q
Multi-Head Cross-Attention
Dense
Stable-Gated
Sigmoid
Dense
A
B
Feature: price × news × macroeconomic indices
Temporal aggregation
(+)
Feature aggregation
Cross-Entropy Loss
UP
FLAT
DOWN
t: trading day, prediction for day t+1

📍 **8 stock tickers:** TSLA, AAPL, AMZN, MSFT, GOOGL, META, BA, WMT

📅 **Period:** 01/01/2022 – 23/06/2025 **(869 days)**

📈Price data

Yahoo Finance (yfinance)

Provides quantitative measures of daily market behavior.

- **Information:** OHLC (open, high, low, close prices).
- **Role:** Reflects short-term price trends and actual trading dynamics.

📰 Financial news

Total scale: **64,894 articles**

Alpaca News API

A financial infrastructure platform specialized for algorithmic trading in the United States.

- **News type**: Real-time ticker-specific news from partners such as Benzinga and Reuters, including earnings reports, mergers, and related events.

The Guardian (Business)

A globally recognized UK-based newspaper with a history of more than 200 years.

- **News type**: In-depth analysis of market context and multi-perspective viewpoints.

Preprocessing and unified dataset construction workflow

Branch I – Price indicators

- **Yahoo Finance**
  8 tickers – daily OHLC
- **Volatility-adjusted log returns with** rolling window
- **Z-score normalization**
  (training set only)
- **Percentile-based labeling**
  DOWN / FLAT / UP
- **Tensor (B, T, 3)**

Branch N – Financial news

- **Alpaca + Guardian**
  64.894 articles
- **Temporal alignment and deduplication**
  (16:00 ET · 3 filters)
- **LLM-based triple extraction**
  (Gemini Flash · S-R-O)
- **FinBERT encoding**
  (768 dimensions · weighted average)
- **Tensor (B, T, 768)**

Branch M – Macroeconomic indices

- **Yahoo + FRED**
  5 macroeconomic indices
- **Yield spread calculation**
  (DGS10 - DGS2)
- **1-day lag + missing-value handling**
  Causal · Forward fill
- **Z-score normalization**
  (training set only)
- **Tensor (B, T, 5)**

Unified dataset
869 trading sessions · 8 tickers

Prompt Engineering & Schema

Raw Article

TYPE A
Earning / EPS

TYPE B
M&A / Corporate

TYPE C
Legal / Regulatory

TYPE D
Opinion / commentary

→

Gemini 2.0 Flash
3-tier prompt
+JSON schema

→

Triplets (S - R - O)

```
EXTRACTED TRIPLE (S-R-O)
-------------------------------------
Subject: COMP - "Apple"
Relation: ANNOUNCES
Object: PRODUCT - "iPhone"
-------------------------------------
Subject: COMP - "Apple"
Relation: SIGNALS
Object: ECON_ID - "EPS $2.54"
```

8 entity types:
COMP, PERSON,...

14 relation types:
ANNOUNCES, NEG_IMPACT,...

Scores per triple:
confidence, price_impact, relevance

Macroeconomic Indicators

Indicator	Ticker & Source	Economic Significance
CBOE Volatility Index	**^VIX** Yahoo Finance	Measures market-implied volatility expectations from options; gauges investor “fear.”
Treasury Yield Spread	**DGS10 - DGS2** FRED (Fed Reserve Data)	Difference between 10-year and 2-year Treasury yields; reflects business cycles and recession risks.
Broad Market Index	**^GSPC** Yahoo Finance	Daily performance of the S&P 500 Index; captures overall market momentum signals.
US Dollar Index	**DX-Y.NYB** Yahoo Finance	Measures the relative strength of the USD against major global currencies; impacts multinational firms.
WTI crude oil price	**CL=F** Yahoo Finance	West Texas Intermediate futures prices; proxies global energy cost shocks and inflation pressures.

DOWNSTREAM PIPELINE

All variables strictly undergo **Rolling Z-Score Normalization** post-lag alignment to achieve uniform scaling, followed by a **Linear Projection Layer** to map features into a consistent latent dimension for cross-modal fusion.

BASELINE MODELS: MODERN & HIGH-PERFORMANCE

Benchmarking NKTrif Against State-of-the-Art Peer-Reviewed Architectures

GROUP 1: PRICE ONLY — FOUDATION

LSTM - 2018

Standard 2-layer Long Short-Term Memory network for temporal dependencies.

EUROPEAN JOURNAL OF OPERATIONAL RESEARCH

ALSTM - 2018

LSTM with temporal attention to capture informative historical sessions.

IJCAI / / FENG ET AL.

GROUP 2: PRICE +NEWS — MULTIMODAL

ALSTM-W - 2018

Window-averaged FinBERT embeddings concatenated with technical price data.

ESTABLISHED NEWS BASELINE

SOTA COMPARISON

SLOT-2022

Cross-projection integration between price and news signals. Top-performing peer.

IEEE BIGDATA / / SOUN ET AL.

GROUP 3: PRICE + MACRO — SYSTEMIC

DA-RNN - 2017

Dual-stage attention for systemic feature selection

IJCAI / I QIN E T AL.

DTML 2021

Transformer architecture capturing multi-level market

ACM KDD / / YOO ET AL.

RECENT

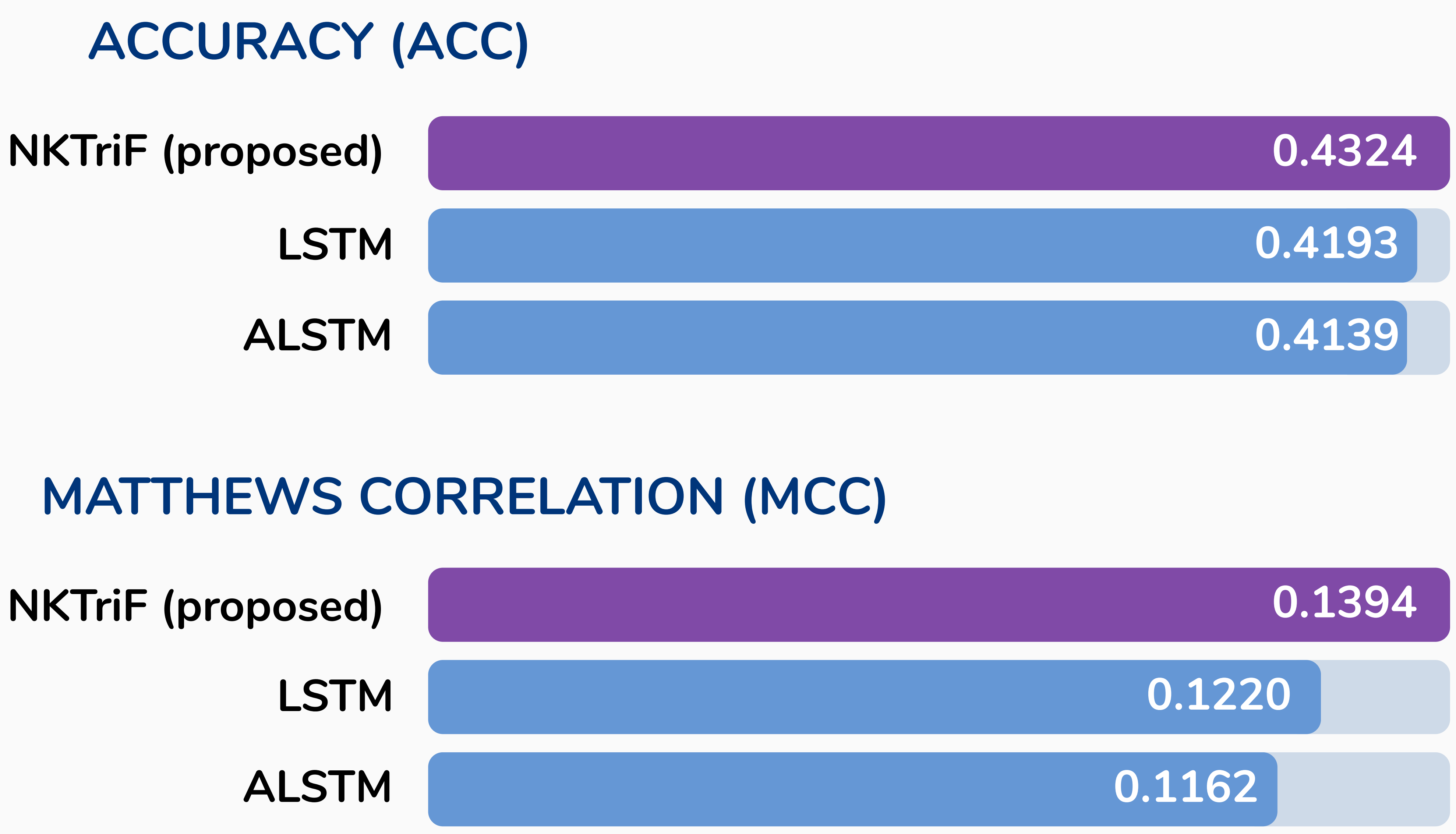
ESTIMATE

Latest baseline using multi-head attention for late-fusion intergration

ACM WSDM / / HUYNH ET AL.

Comparison with Group 1 – Price only

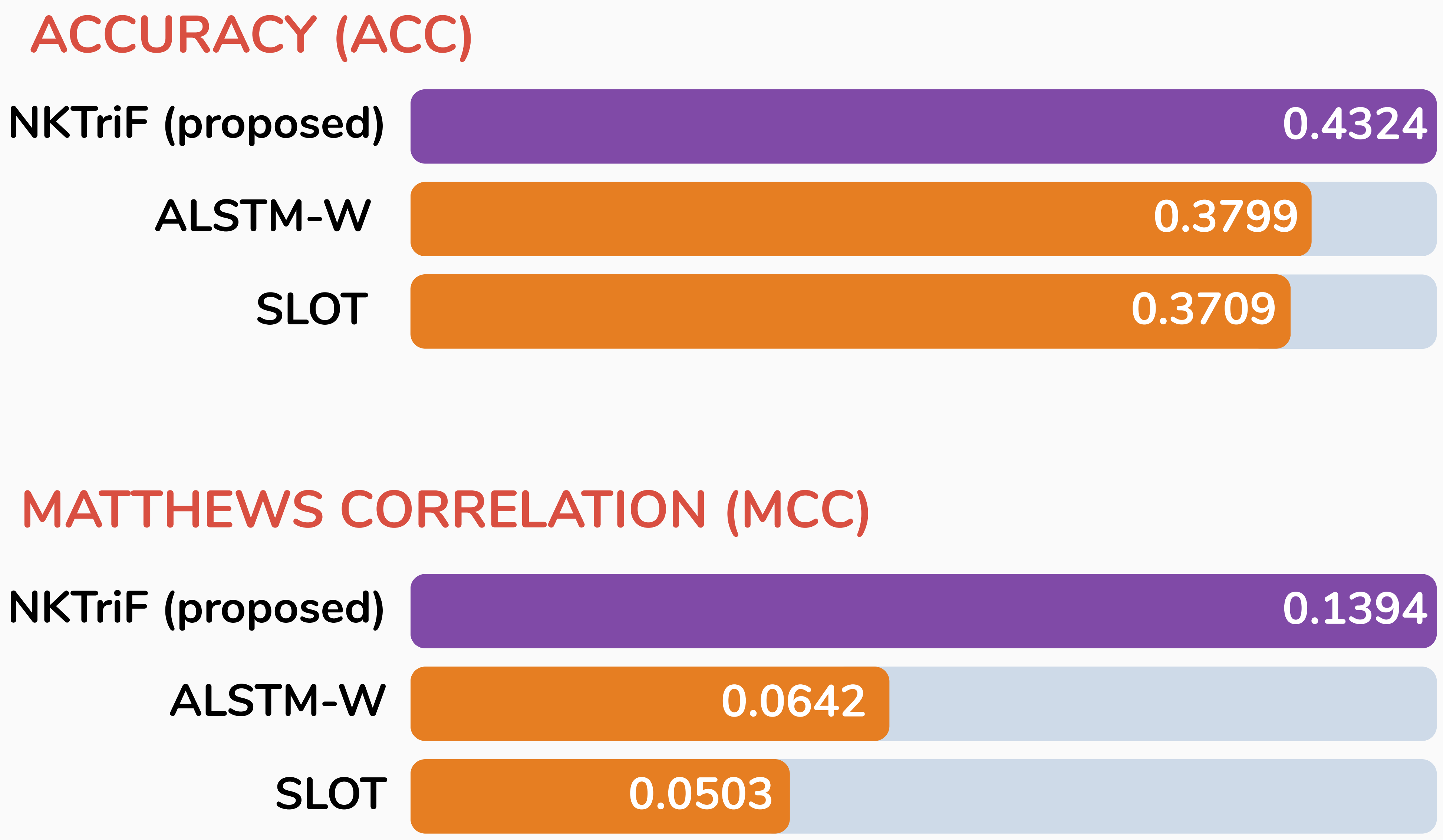
Performance compared with price-only models



- NKTriF leads on both ACC and MCC across all price-only baselines.
- The gap is structural: price-only models **cannot respond promptly** to unexpected market-moving events.

Comparison with Group 2 – Price + news

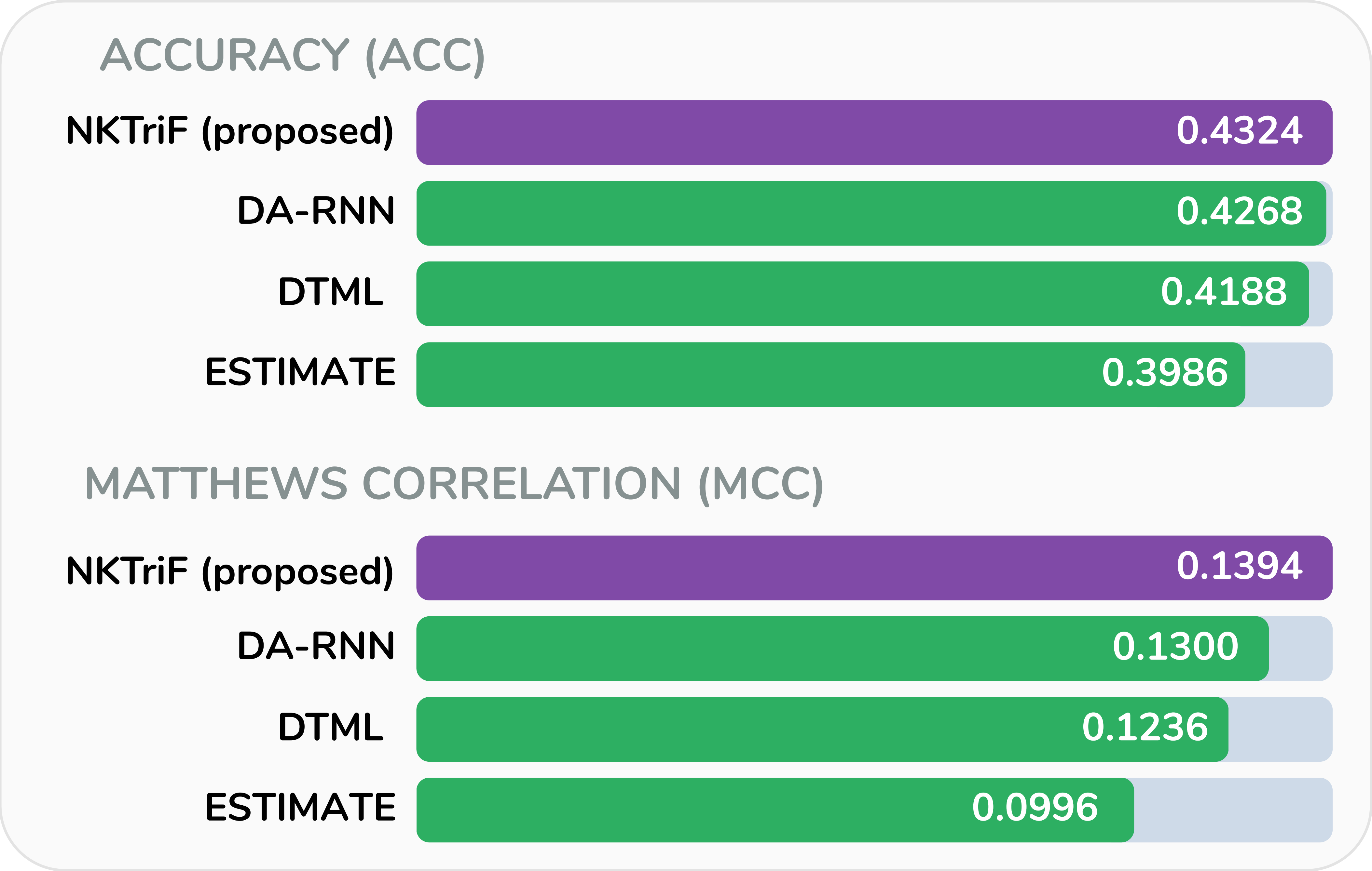
Performance compared with Financial news-based models



- NKTriF leads on both metrics — and the MCC gap is nearly 3x against SLOT
- Notably, both ALSTM-W and SLOT perform worse than price-only baselines. Adding news without proper control doesn't help — it hurts.
- NKTriF outperforms these models by **structuring news into knowledge triples** to reduce textual noise.

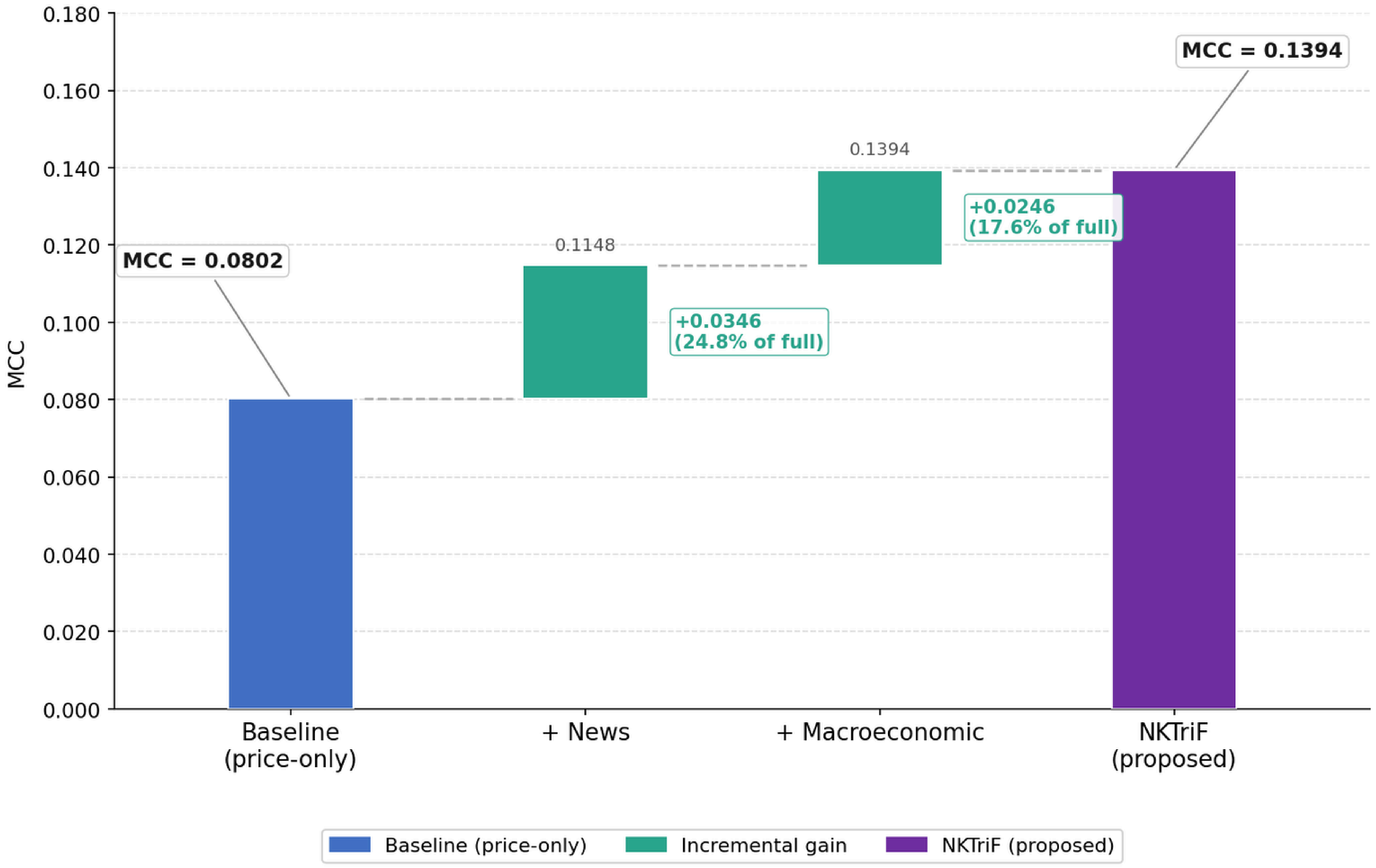
Comparison with Group 3 – Price + macroeconomic

Performance compared with Macroeconomic Index-based models



- NKTriF leads, but DA-RNN follows closely — it's the strongest competing baseline in the entire benchmark.
- macroeconomic information is genuinely predictive, and models that incorporate it perform better.

Modality contribution analysis (Ablation study)

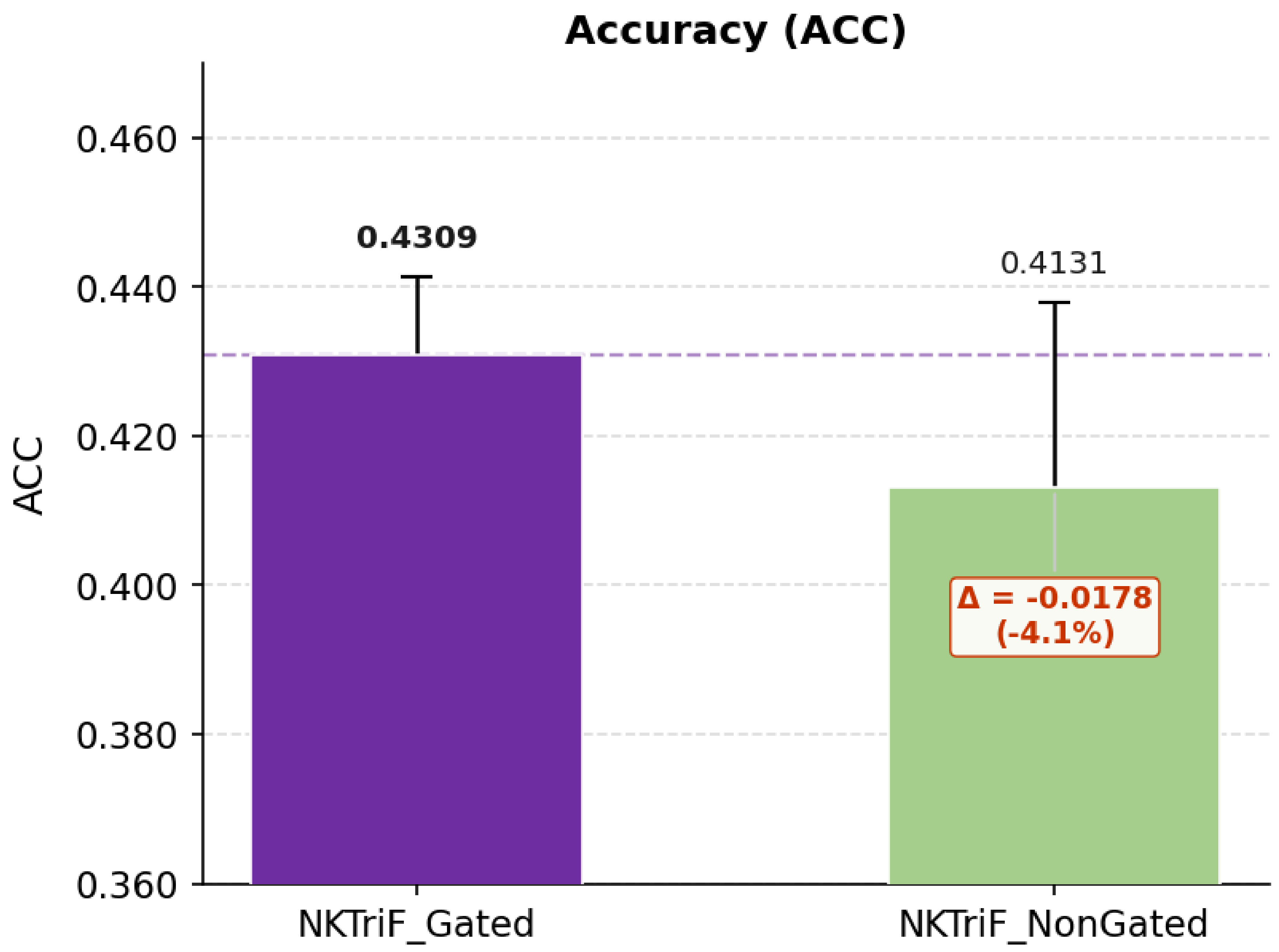


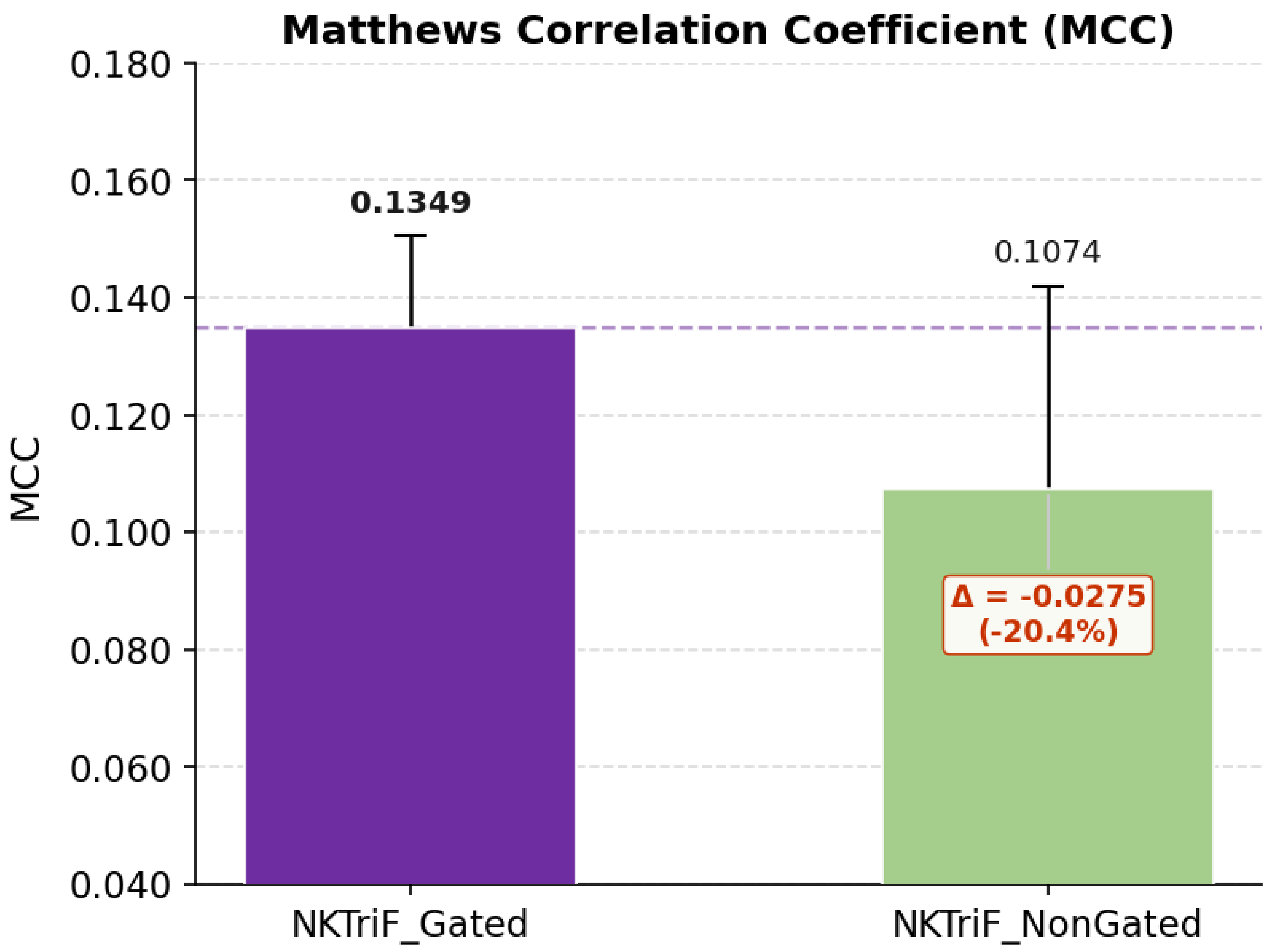
- Both modality data contribute meaningfully to prediction performance.
- **news provide direct event signal**
- **macro stablilize the model variance**
-

Standard deviation (Std) of MCC

- Full model: 0.0152
- Without macro: 0.0321 **(2.1× higher)**

Role of the Gated fusion mechanism (Gated VS No_Gated)



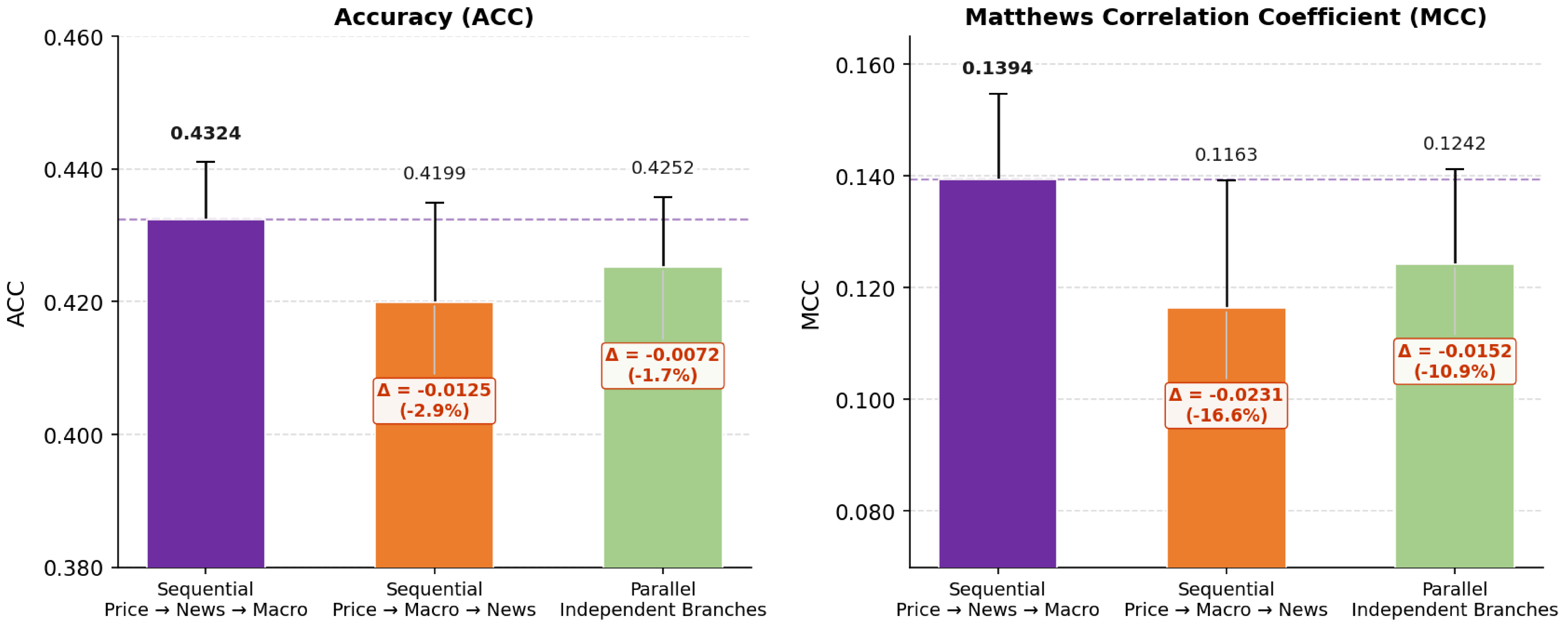


ΔACC = −0.0178 (−4.1%)

ΔMCC = −0.0275 (−20.4%)

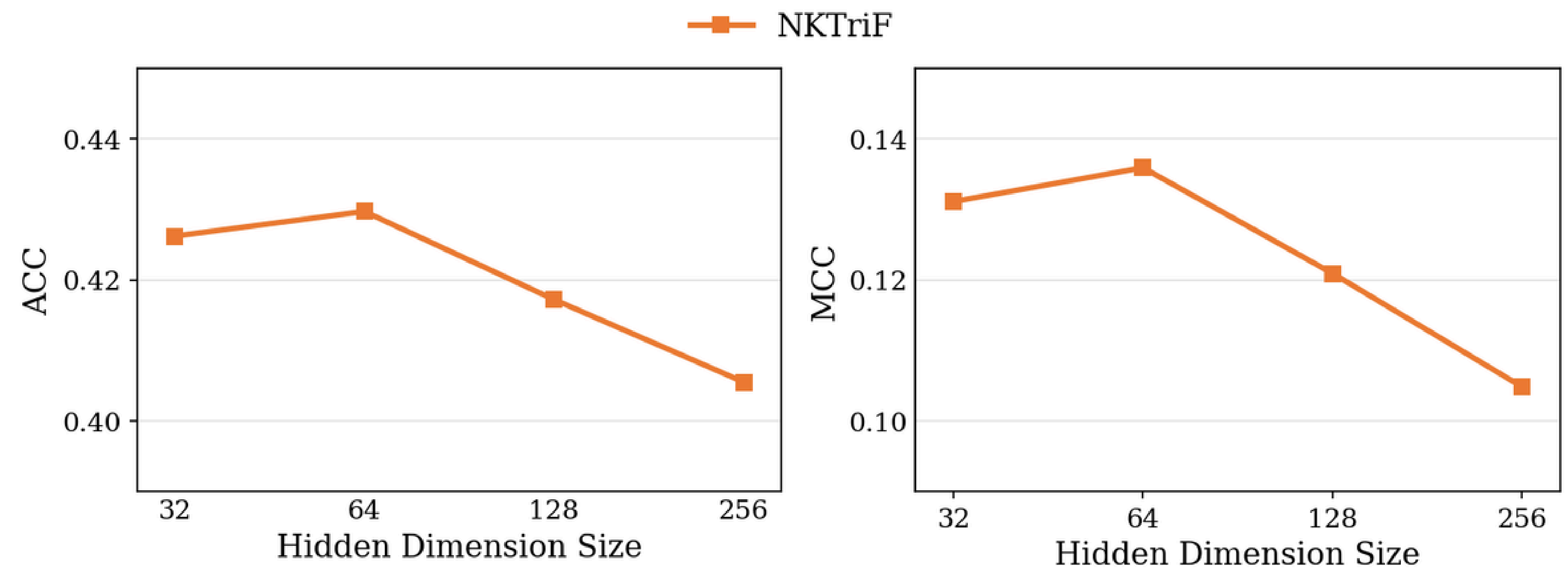
- Removing the gate hurts performance on both metrics
- More importantly, the gate reduces model variance, making predictions consistent across different market conditions

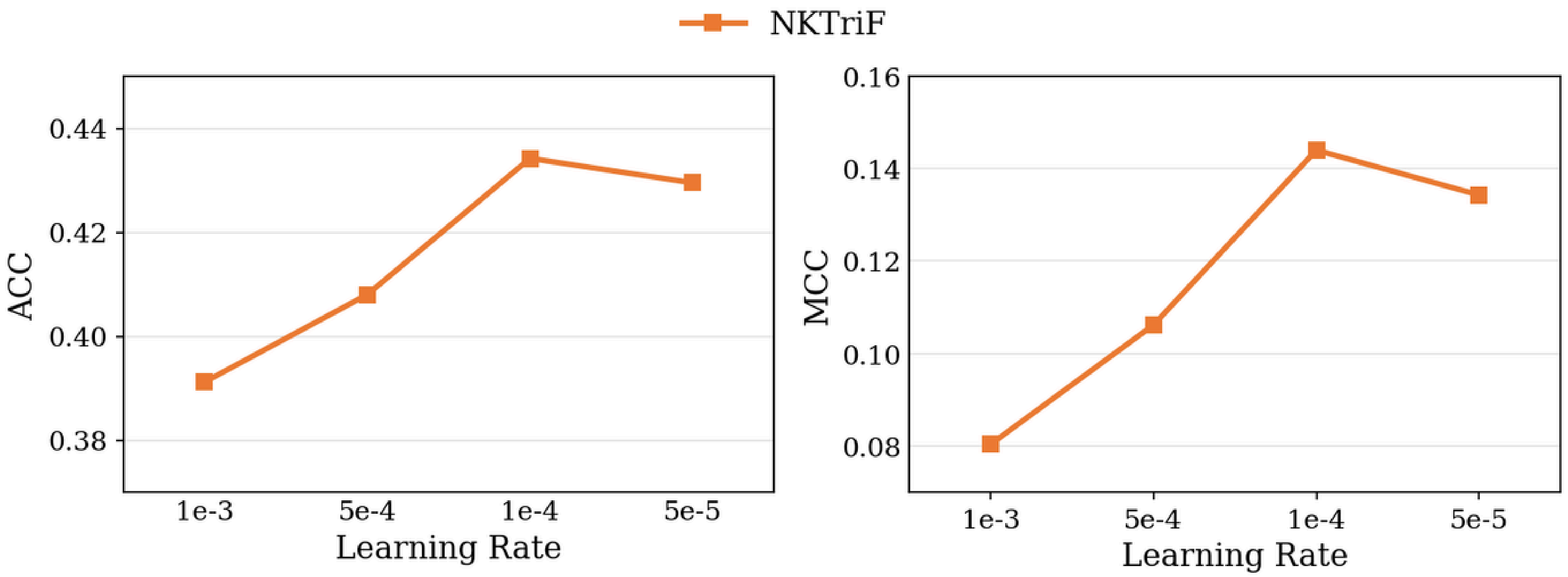
Role of fusion order across data sources

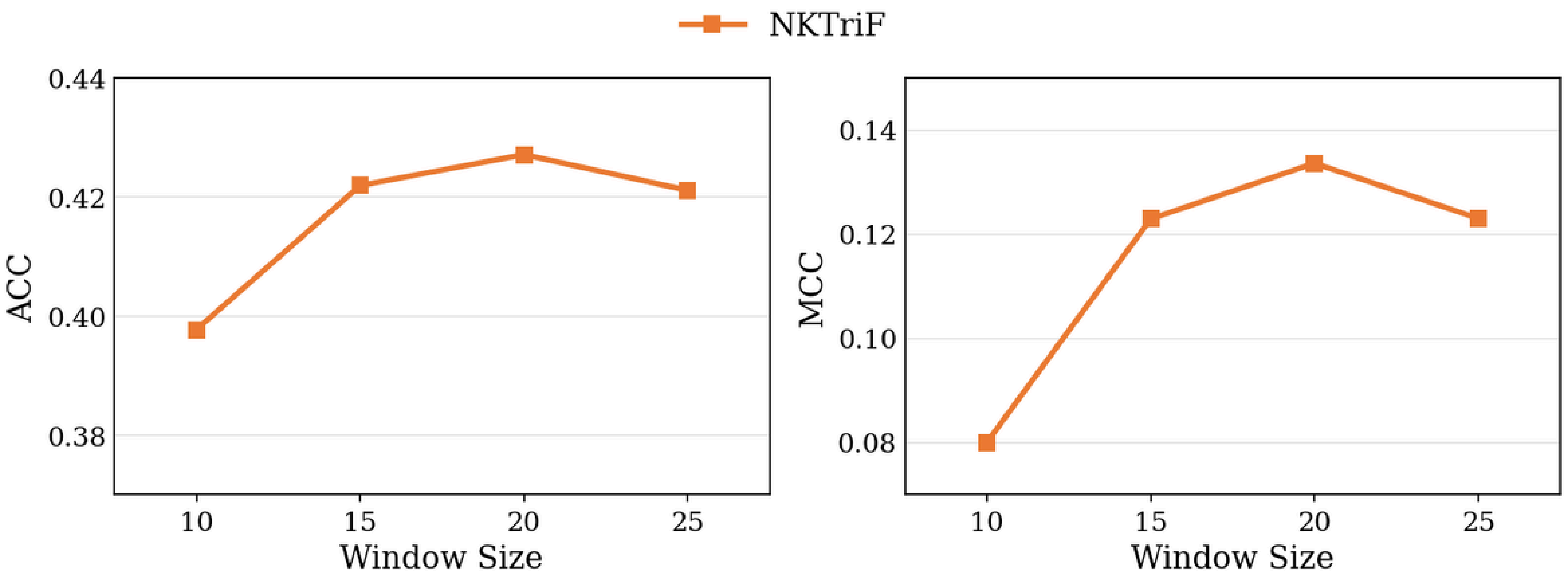


- **News:** carries short-horizon directional event signals, making it suitable for early integration into the price representation.
- **Macro:** reflects background market context and serves as a second-stage refinement of the representation.

Hyperparameter Sensitivity Analysis

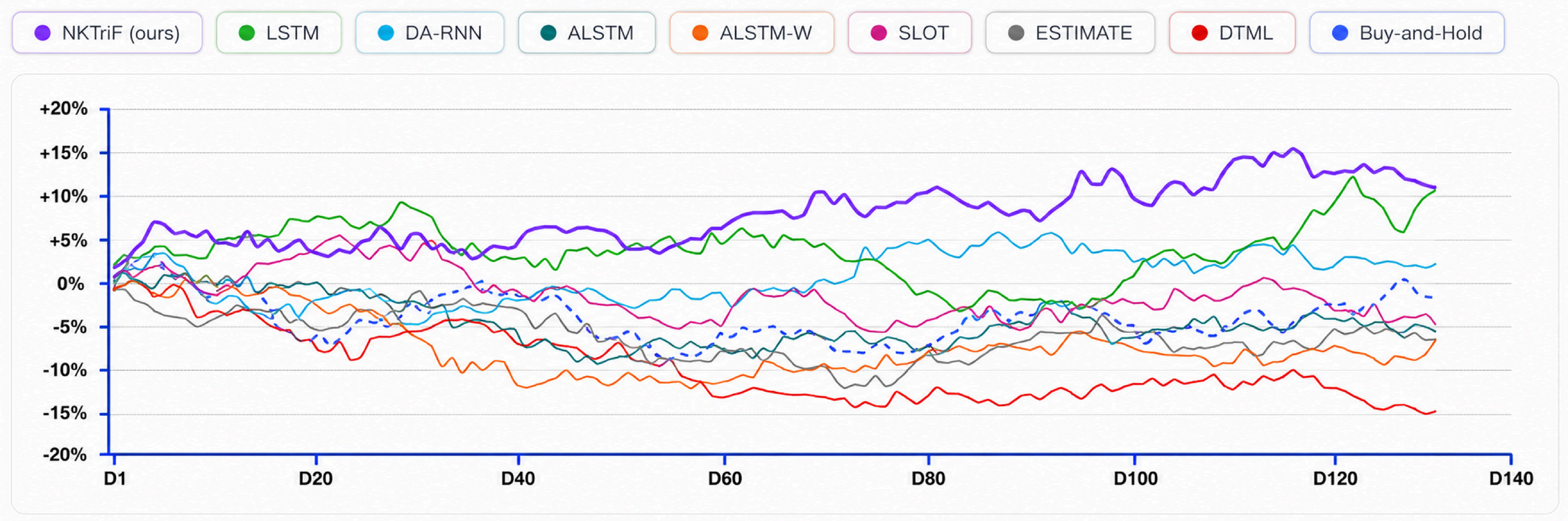






Hyperparameter sensitivity analysis

Trading simulation



NKTriF

0.85
Highest Sharpe Ratio

+27.68%
Best Annual Profit

+0.7%
Better profit than LSTM

01 Main Contributions

01 **Developed NKTriF** to integrate price, news, and macro signals through price-anchored two-stage Gated Cross-Attention.

02 **Achieved ACC = 0.4324** and MCC = 0.1394, exceeding DA-RNN by 1.3% ACC and 7.2% MCC.

03 **Structured raw financial news** into entity-relation-object triples to preserve stock-specific event signals.

02 Future work

- **Build a Vietnamese financial news** pipeline with company-to-ticker resolution.
- **Adapt Vietnamese financial language** models for entity-relation-object extraction.
- **Develop a periodic disclosure encoder** for earnings reports, filings, and earnings calls.

THANK YOU!

Q&A section

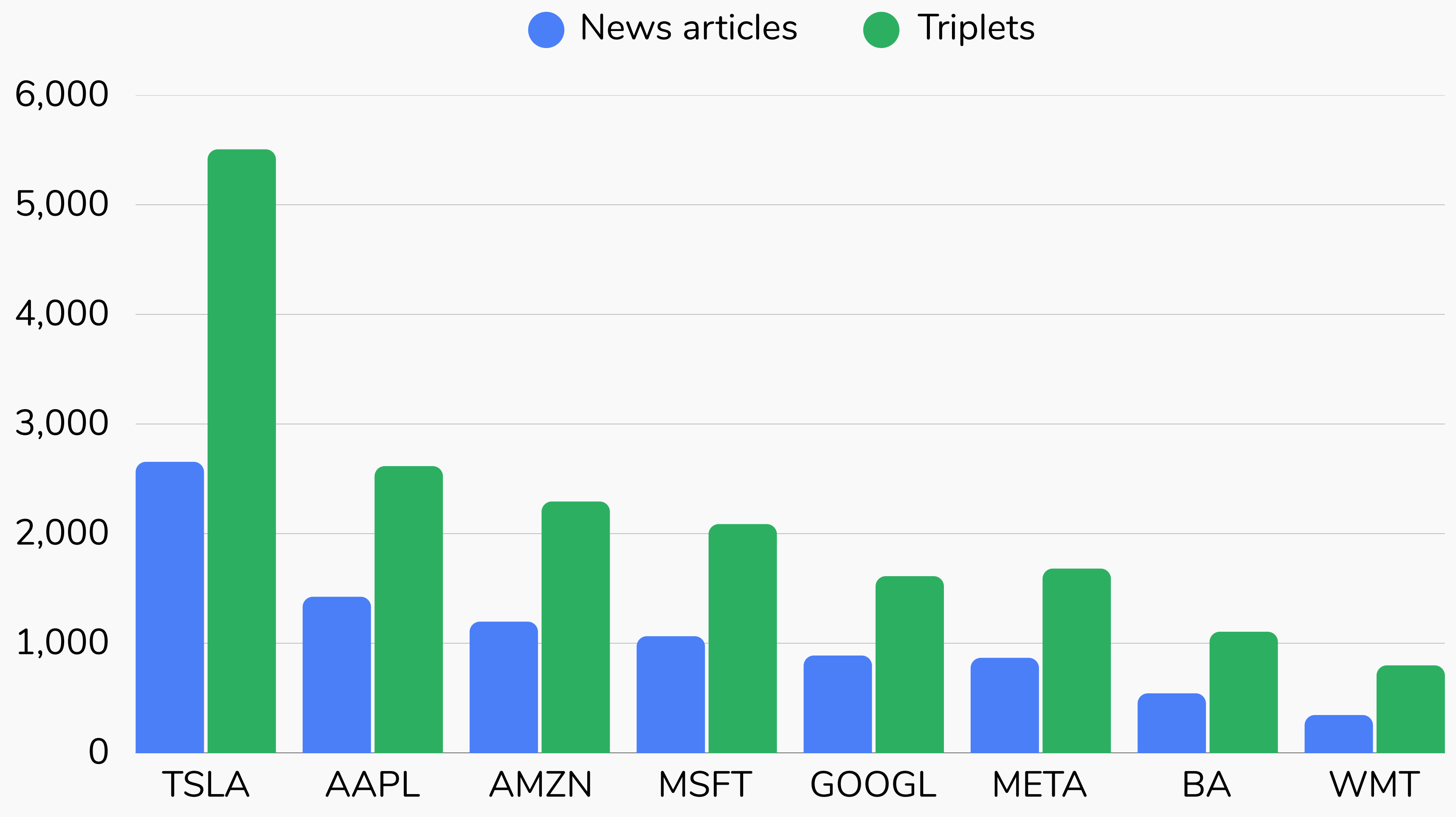
NKTRIF - A MULTIMODAL STABLE FUSION APPROACH FOR FINE-GRAINED STOCK MOVEMENT PREDICTION

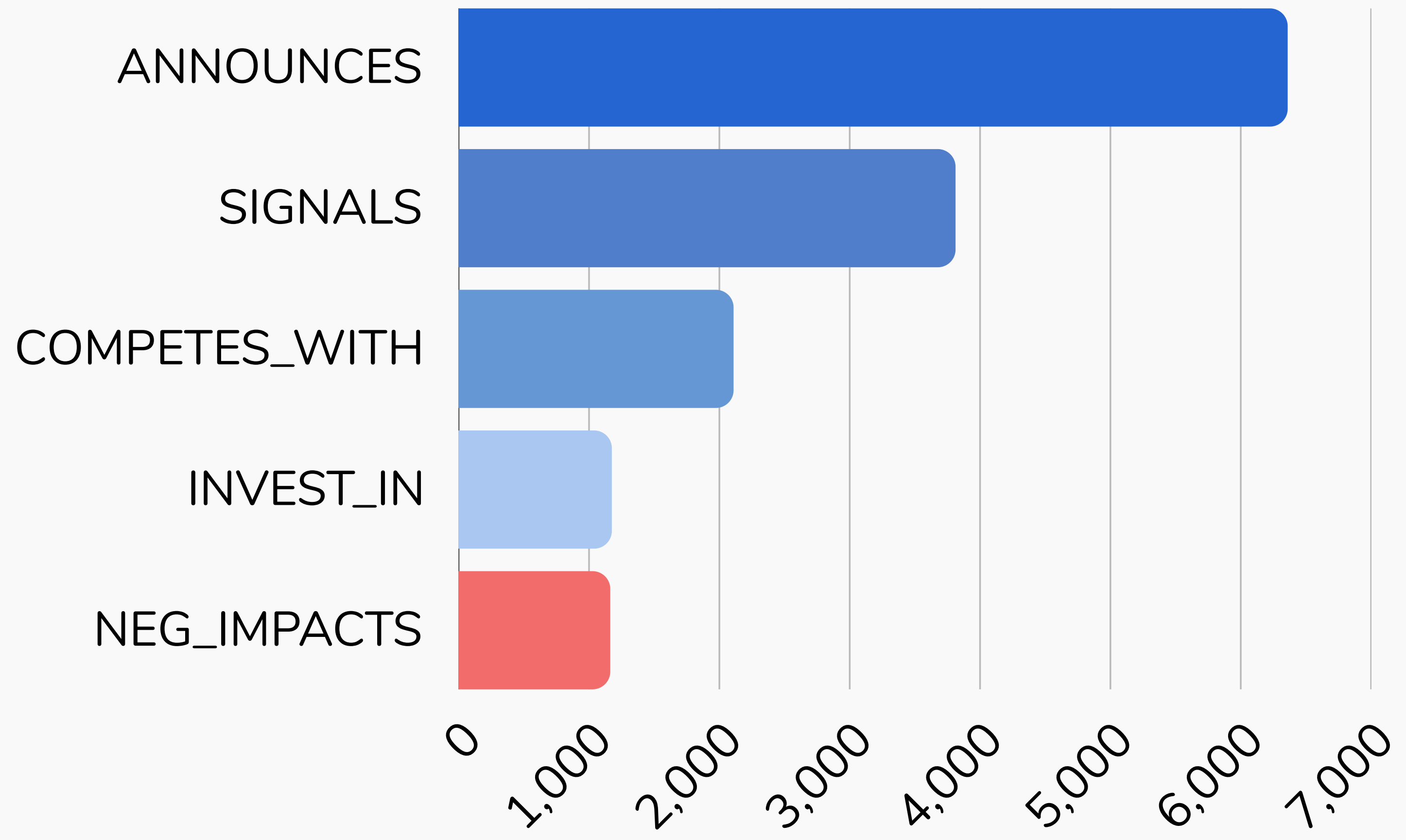
Extraction quality: compression & signal preservation

94:1
Compression ratio
516 tokens/article → 5.5 tokens/triple

17,698
Structure
Ectracted 8,983 valid article-ticker pairs

1-2
Triplets per article
Focused event-level distillation





Trading simulation

129 - Day Test Period (Backtest Horizon)

8-Asset S&P 500 Ticket Universe

4 bps/side Transaction Costs (Net of Friction)

Decision Signal

$$Score_{i,t} = P_{UP} - P_{Down}$$

Converts 3-class probabilities into a directional commitment signal.

→ Filter: The "FLAT" class dampens signals in low-conviction states.

→ Range: Continuous score from -1 (Bearish) to +1 (Bullish).

Portfolio Strategy

$$w_{i,t}^{+} = \frac{1}{2} \cdot C \cdot \frac{Score_{i,t}}{\sum_{j \in L_t^{+}} Score_{j,t}}, \quad \forall i \in L_t^{+}$$

$$w_{i,t}^{-} = \frac{1}{2} \cdot C \cdot \frac{|Score_{i,t}|}{\sum_{j \in L_t^{-}} |Score_{j,t}|}, \quad \forall i \in L_t^{-}$$

Market-Neutral: Equal capital in long and short legs

Confidence-Weighted Allocation

Annualized Sharp Ratio

$$Sharpe = \frac{\bar{r}}{\sigma(r)} . \sqrt{252}$$

Quantifies risk-adjusted efficiency across the test horizon.

Annualized Sharp Ratio

$$\left(\prod_{t=1}^{T} \left(1 + r_t\right) \right)^{252/T} - 1$$

Measures the final compounded growth rate.

Knowledge extraction pipeline from financial news

Step 1

Knowledge triple extraction

- Gemini 2.0 Flash + three-tier prompt (system · few-shot · JSON Schema)
- Output: subject, relation, object
- With confidence, price_impact_score, relevance_to_ticker

Step 2

Triple filtering and cleaning

- Remove triples with confidence < 0.65 and relevance < 0.50
- Semantic deduplication
- Entity name normalization · Relation direction correction

Step 3

Semantic encoding (ProsusAI/FinBERT)

- Convert each triple into a short event sentence
- Encode with ProsusAI/FinBERT → 768-dimensional vector ([CLS] token)
- Aggregate into a daily vector using a weighted average (prioritizing high-confidence triples)

Human Evaluation